Math 1200 Lecture 17

Nov 3 2026

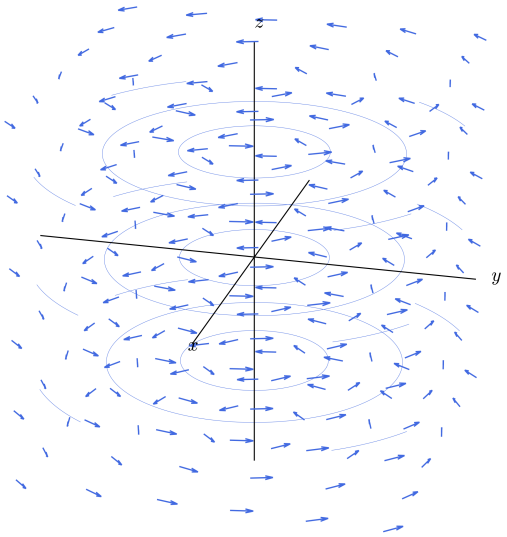
Topics: 16.4-16.5: Green's theorem (more examples), curl Name:_____
Collaborators:_____

Group work 1 Let $f(x, y, z) = xy + 3x\sin(z) + z^2$. What do we expect is the value of $\text{curl}(\nabla f)$? Calculate to verify your expectation.

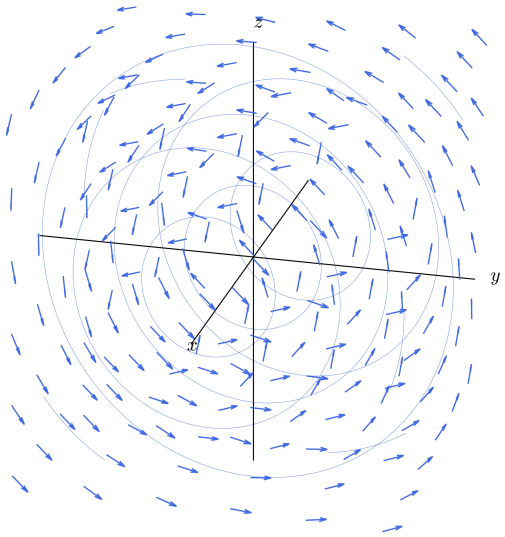
Group work 2 (Discussion) Let $\vec{F_{2d}} = \langle P, Q\rangle$ be a 2d vector field. This produces a vector field $\vec{F} = \langle P, Q, 0\rangle$ in 3d with all vectors parallel to the xy-plane. How does the vector curl $\nabla \times \vec{F}$ relate to the scalar curl $Q_x - P_y$?

Group work 3 Draw the vector fields $\text{curl}(\vec{F}) = \nabla \times \vec{F}$ overlaying the fields $\vec{F}$.

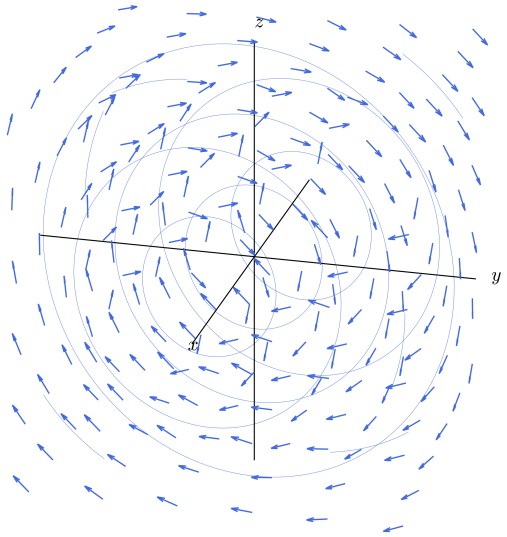
$\vec{F} = \langle -y,\, x,\, 0 \rangle$



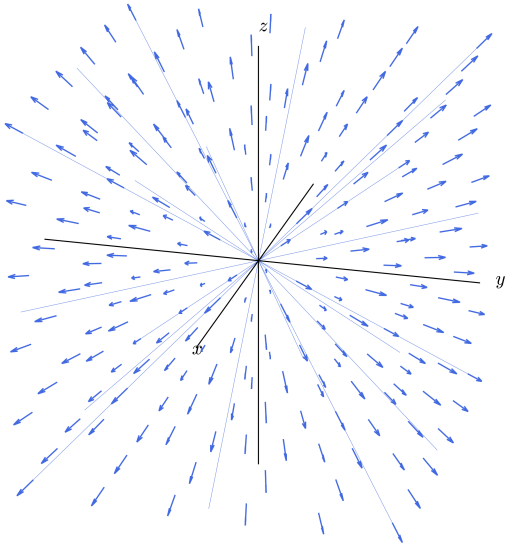
$\vec{F} = \langle z-y,\, x-z,\, y-x \rangle$



$\vec{F} = \langle y-z,\, z-x,\, x-y \rangle$



$\vec{F} = \langle 2x,\, 2y,\, 2z \rangle$



Group work 4 Calculate the following vector line integrals

a) $\int_C \langle e^{y^2} - 1, 2xye^{y^2} + x\rangle \cdot d\vec{r}$ when C is the path from $(-1, 0)$ to $(0, 1)$ followed by the path from $(0, 1)$ to $(1, 0)$.

b) Let C be the curve parametrized by $\vec{r}(t) = \langle \cos(t), \sin(t), \cos(t)\rangle$ for $0 \leq t \leq 2\pi$ and $\vec{F} = \langle 2x, 2y, 2z\rangle$. Find $\int_C \vec{F} \cdot d\vec{r}$.